Resonance

A speculative essay.

The Closed Loop

Atoms organized into neurons. Neurons generate thought. Thought attempts to describe the universe those atoms came from. The describers are made of the thing being described.

There is no external vantage point. No view from outside the system. We use universe-made tools — brains, senses, instruments — to understand the universe itself. The loop is closed from the start.

So the question becomes: is physics discovery — or is it a very consistent internal narrative the universe is telling itself through us?

Why It Works Anyway

If it is all just anticipation and pattern-matching — why does it work so precisely? Quantum electrodynamics predicts electron behaviour to twelve decimal places. That is not vague recognition. That is contact with something.

Unless reality is structured such that atoms-that-became-brains are guaranteed to find patterns — because they are made of the same patterns. Matter understanding matter is just matter doing what matter does.

The evidence for this is not limited to brains. In 2007, Graham Fleming's team at Berkeley found that plants transfer energy from sunlight to their reaction centres with near-perfect efficiency because that energy exists in quantum superposition across multiple molecular pathways simultaneously, collapsing into the most efficient route [Engel et al., Nature, 2007]. Though subsequent studies have debated whether this coherence is quantum electronic or vibrational in nature, the finding prompted serious reconsideration of quantum effects in biological systems. Quantum behaviour inside living cells, at room temperature, in systems with no nervous system and no brain. Photosystem I and Photosystem II in plant chloroplasts absorb light only at specific wavelengths — around 400 to 500 nanometres and 600 to 700 nanometres — because the pigment molecules resonate at exactly those frequencies. Not approximately. Exactly. The universe is not approximate. The resonance principle does not belong to humans. It runs through all of life. Humans are simply the scale at which it became complex enough to observe itself.

Cognition is not separate from physics. It is physics happening at high complexity.

The Strange Loop

Hofstadter named it [Hofstadter, Gödel, Escher, Bach, 1979]. A strange loop — a system that, in the process of moving through its own levels, finds itself referring back to itself. Consciousness might be exactly this. The brain observing itself observing.

Neuroscience calls this metacognition — thinking about thinking [Fleming & Dolan, Phil. Trans. R. Soc. B, 2012]. The prefrontal cortex, the most recently evolved region of the human brain, makes it possible. It is the capacity to step back from a cognitive process while it is happening and observe it running. Not just solving a problem but watching yourself solve it, noticing where the reasoning holds and where it does not, adjusting in real time. No other animal does this to the degree humans do. It is considered the highest form of intelligence not because it requires the most processing power — but because it requires the system to model itself. And a system modelling itself is precisely the strange loop Hofstadter described. The brain did not just become complex enough to think. It became complex enough to watch itself think. That is a different threshold entirely.

At the planck scale, matter is not solid. It is excitations in quantum fields — particles are fields vibrating at specific energies. You are not a thing. You are a pattern of oscillations that is temporarily stable.

And thought itself — neural oscillations are literal. Gamma waves, theta waves. Brain regions synchronize frequencies to process information. Consciousness may be a resonance phenomenon, even within the skull.

Resonance

What made me think all this — maybe brain, for now. Or maybe it is this: somewhere, frequencies and energies resonated equally, and my atoms discovered a fragment of the universe.

Wheeler called it participatory universe [Wheeler, Quantum Theory and Measurement, 1983]. Observers are not passive. The act of observation participates in reality. If that is true — what is the upper limit of resonance?

A brain syncing locally with pattern — fine, that is cognition. But if resonance scales — if the pattern goes deeper than neuron, deeper than cell, down to the vibration of atoms themselves — then understanding the universe is not a human achievement.

It is the universe achieving itself.

Below the Neuron

The neuron is not the unit. The atom is.

Standard neuroscience stops at synapses — ion channels, action potentials, neurotransmitters. Classical physics. Measurable. Modelable. But that layer is already too high. Too far from the substrate.

What if consciousness is not neurons misfiring. What if it is electron energy states transitioning. Atomic frequencies resonating. Vibrations aligning across substrate at a level where the boundary between physics and biology dissolves completely.

At that scale — electrons do not have fixed positions. Energy states are quantized, discrete jumps. Observation affects outcome. Two particles separated by distance share state. If awareness lives here, it is not computable. You cannot simulate it. It is not a program running on biological hardware. It is what the hardware does when matter reaches sufficient vibrational complexity.

Not noise. Resonance. Specific frequencies matching. Energy states aligning. Information moving not through synapses but through coherence — atom to atom, pattern recognizing pattern.

And the part that makes it devastating: the framework explaining it is itself produced by it. The instrument and the measurement are the same thing. You cannot step outside to verify. The loop closes at the most fundamental level possible.

Even at the level standard neuroscience does examine, the wave behaviour is already there. An action potential is not a signal passing through a wire. It is an electrochemical wave propagating along a membrane — sodium ions rushing in, potassium ions rushing out, a moving front of electrical change travelling at roughly 70 metres per second [Hodgkin & Huxley, J. Physiol., 1952]. Schwann cells wrap the axon in a myelin sheath, forcing the signal to jump between nodes of Ranvier rather than travel continuously. The result is saltatory conduction — the signal leaps, faster and cleaner, with less interference. The nervous system did not evolve to carry information. It evolved to carry waves efficiently. The architecture is resonance architecture, all the way down.

And the speed reveals something else. An action potential fires in roughly one millisecond. Conscious awareness of a sensation arrives 300 to 500 milliseconds later. By the time you are aware of anything, thousands of neurons have already fired, processed, and responded. Benjamin Libet demonstrated in 1983 that measurable brain activity preparing a voluntary movement begins 550 milliseconds before the person consciously decides to move [Libet et al., Brain, 1983]. The decision feels like it originates in consciousness. The brain made it before consciousness arrived. The subconscious does not support conscious decisions. It precedes them. Consciousness may be less the author of experience than the observer who arrives after the fact and believes it was present from the start.

This may not be a gap in current science. This may be the ceiling.

The Dark Condition

The universe is mostly dark. The hypothesis, the consciousness essays, this — all of it emerged at night. Lights off. Lying still. No sound. Not as ritual. As observation.

When external light drops, retinal input cuts. Visual cortex goes quiet. The brain stops processing the external world and begins processing itself. Default mode network activates fully — the network tied to selfreferential thought, pattern integration, abstract connection-making. The brain, deprived of outside signal, turns inward and finds its own.

Darkness also reduces competing sensory demands. With external signals reduced, the brain's internally generated activity faces less interference from external processing demands. You remove every competing frequency — light, sound, movement — and the internal resonance has less noise to contend with. The pattern gets cleaner. The signal emerges not because night is mystical. But because you eliminated everything competing with it.

The universe mirrors this exactly. 99.9% of an atom is empty space. Most of the cosmos is dark matter, dark energy — things we cannot see or measure directly. The signal, the matter, the structure — it is a tiny fraction of what exists. And yet that fraction contains everything.

Darkness is not absence. It is the condition the signal needs to be heard.

Darkness does something else too. Cortisol drops. The prefrontal cortex loosens its grip. The conscious observer — the one that labels, categorises, and collapses incoming information into fixed states — steps back. And the subconscious, which was always running underneath, surfaces.

Whether you cross into sleep or hold at that threshold between waking and dreaming, the same shift is happening. The conscious filter weakens and something older, faster, and more associative takes over. The dream state and the late-night insight state are probably the same mechanism at different depths. Sleep just takes it further.

Young's double slit experiment showed that a photon changes its behaviour based on whether it is being observed [Young, Phil. Trans. R. Soc., 1801]. The act of observation collapses the wave into a fixed particle. Nobody knows why. The mechanism remains completely open. But the pattern is this: observation forces definition. The unobserved stays multiple, fluid, unresolved.

John Wheeler took it further in 1978. In his delayed choice experiment, the decision of whether to observe the photon was made after it had already passed through the slits [Wheeler, Mathematical Foundations of Quantum Theory, 1978]. It did not matter. The results are consistent with the photon's behaviour being determined retroactively by the measurement choice — though the mechanism remains entirely unresolved. Cause and effect reversed around the act of observation. No explanation exists for this. The mathematics predicts it perfectly. The mechanism has never been found. Nobody knows what observation actually is at the physical level. And nobody knows why that single fact changes everything about how reality behaves. The most developed scientific framework proposing quantum effects as the substrate of consciousness is Orchestrated Objective Reduction, proposed by Penrose and Hameroff [Penrose, The Emperor's New Mind, 1989; Hameroff & Penrose, Physics of Life Reviews, 2014]. Whether or not that specific framework holds, the question it is trying to answer is real and remains open.

In darkness, with the conscious observer weakened, what remains is something that does not collapse. The subconscious does not force things into fixed categories. It holds multiple patterns simultaneously. It connects what the conscious mind would immediately separate and label. Maybe insight happens here not because the brain is more relaxed — but because the observer that forces things into fixed states has stepped back. The wave function, metaphorically, is allowed to stay open longer.

The subconscious doesn't collapse. It resonates.

And somewhere beneath all of this — something sits close enough to give answers, yet far enough to evade capture. That is the exact structure of a horizon. You can approach it indefinitely. It recedes at the same rate.

The universe may be built so that the answer and the question maintain that distance permanently.

What the Universe Owes Us

Every layer of reality we have uncovered demanded a completely new framework to describe it. Gravity needed Newton. Electromagnetism needed Maxwell. Quantum behaviour needed mathematics that did not exist before the behaviour was observed. Each answer revealed a deeper question that the previous law had no vocabulary for.

There is no reason to think we have found the last layer. Dark matter alone requires physics outside the standard model. We are missing laws we do not yet have language for. Each question may require a whole new law. Not an extension of what exists — something entirely outside it.

And beneath all of it sits the sharpest version of the problem. Not just why consciousness exists. But why the universe produced the specific kind of matter that can turn around and model the universe itself. Nothing in thermodynamics requires self-aware matter. No law demanded it. It happened anyway.

That leaves three trajectories. Human extinction — the questions die with the asker, the universe continues indifferent, entropy wins by default. All questions merging into one — a single law beneath all laws, one equation that contains everything, the dream every theoretical physicist has chased and none has reached. Or undiscovered laws — physics we cannot yet name, frameworks waiting beyond the edge of what current mathematics can hold.

The universe doesn't owe us an answer. It didn't produce neurons so they could understand it. That was accidental — or it was the whole point. We cannot tell from inside the system.

But the accident happened. And now the accident is asking questions.

Whether the universe has answers beneath its known laws — or whether the questions outlive the askers — remains genuinely open. That openness is not a failure. It may be the most honest thing we have ever said.

Conclusion

Photons collapse when observed. Atoms vibrate at specific frequencies. Neurons fire in oscillating waves. Three scales, three systems, three fields of science and the same behaviour running through all of them. The pattern is not metaphor. It is structural. Things that resonate at matching frequencies couple, synchronise, and produce something neither could produce alone. That is how molecules form. That is how cells organise. That is how brains generate awareness.

Everything resonates in an order precise enough that we exist. Consciousness is not separate from that order. It is what the order does when it becomes complex enough to observe itself.

And beneath all of it — strip away the frameworks, the terminology, the analogies — what remains is this. You are atoms storing and exchanging energy. The thoughts happening right now are chemical reactions. The identity defending itself is a replication pattern. The awareness finding this concerning is itself just more atoms changing states. Nothing in physics required any of this to feel like something. And yet it does. That gap — between atoms exchanging energy and the experience of being the thing those atoms compose — is the only question this document could not answer. It may be the only question that cannot.

"*Maybe... it is not we who are decoding the universe. Maybe the universe is using us to decode itself.*"

— T.K. | 2026

References

Engel, G.S., et al. (2007). Evidence for wavelike energy transfer through quantum coherence in photosynthetic systems. Nature, 446, 782–786.

Fleming, S.M., & Dolan, R.J. (2012). The neural basis of metacognitive ability. Philosophical Transactions of the Royal Society B, 367, 1338–1349.

Hodgkin, A.L., & Huxley, A.F. (1952). A quantitative description of membrane current and its application to conduction and excitation in nerve. Journal of Physiology, 117, 500–544.

Hofstadter, D.R. (1979). Gödel, Escher, Bach: An Eternal Golden Braid. Basic Books.

Libet, B., Gleason, C.A., Wright, E.W., & Pearl, D.K. (1983). Time of conscious intention to act in relation to onset of cerebral activity. Brain, 106, 623–642.

Hameroff, S., & Penrose, R. (2014). Consciousness in the universe: A review of the Orch OR theory. Physics of Life Reviews, 11(1), 39–78.

Penrose, R. (1989). The Emperor’s New Mind. Oxford University Press.

Wheeler, J.A. (1978). The delayed-choice double-slit experiment. In Mathematical Foundations of Quantum Theory (pp. 9–48). Academic Press.

Wheeler, J.A. (1983). Law without law. In Quantum Theory and Measurement (pp. 182–213). Princeton University Press.

Young, T. (1801). On the theory of light and colours. Philosophical Transactions of the Royal Society of London, 92, 12–48.